

BOND ORIENTATIONAL ORDER IN LIQUIDS AND AMORPHOUS SOLIDS

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Abstract

Recent theoretical work¹⁾ on the Kosterlitz–Thouless²⁾ model of dislocation-mediated melting in two dimensions suggests the existence of a new “hexatic” phase of matter, intermediate between a solid and a liquid. Hexatic liquids display persistent correlations in the “bonds” joining near neighbor atoms. Although these unusual materials have some properties in common with a liquid crystal, there need not be an anisotropy in the constituent particles. Although computer simulations have produced conflicting evidence for an equilibrium hexatic phase³⁾, the discovery of a “stacked hexatic” phase in bulk smectic liquid crystals has been reported recently⁴⁾.

It is interesting to consider bond orientational order in other contexts. Translational and orientational order can be studied, for example, in randomly packed planar array of hard spheres with two different sizes⁵⁾. For an appropriate size ratio, and a dilute concentration of large spheres, one finds a phase without translational order, but with very long range correlations in the orientations of local hexagonal axes. The phase appears to be quenched analogue of the equilibrium hexatic phase. The peculiar properties of this hexatic “glass” are attributed to the tendency of a dilute array of large spheres imbedded in a medium of smaller spheres to trap dislocations. Significant orientational correlations also appear in more concentrated mixtures, giving rise to a six-fold modulation of the structure function in large, but finite, samples.

Bond orientational order can assume a variety of forms in bulk materials. Crystals disordered by an equilibrium concentration of unbound dislocation loops display bond orientational order with a cubic symmetry⁶⁾. Another possibility is orientational order with an “icosahedral” symmetry. It is well known that an icosahedral clustering of twelve atoms about a central sphere is energetically preferable to crystalline packings for most simple pair potentials⁷⁾. These questions have been studied recently via computer simulations of 864 particles interacting through a Lennard–Jones pair potential⁸⁾. Long-range orientational fluctuations appear upon supercooling about ten percent below the equilibrium melting temperature. The fluctuations suggest a broken

icosahedral symmetry with extended correlations in the orientations of local icosahedral packing units.

More a detailed review of the material sketched above may be found in ref. 9 below.

References

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